
Searching demo

#	LeetCode	Difficulty	Concept
1	Two Sum (#1)	Easy	Linear Search + Hashing
2	Find Numbers with Even Number of Digits (#1295)	Easy	Array Traversal
3	Richest Customer Wealth (#1672)	Easy	Linear Traversal
4	Check if N and Its Double Exist (#1346)	Easy	Search
5	Find the Highest Altitude (#1732)	Easy	Prefix Sum + Search
6	Maximum Number of Words Found in Sentences (#2114)	Easy	Linear Scan
7	Count Negative Numbers in a Sorted Matrix (#1351)	Easy	Linear Search
8	Number of Good Pairs (#1512)	Easy	Nested Search
9	Find Pivot Index (#724)	Easy	Sequential Search
10	Valid Mountain Array (#941)	Easy	Linear Traversal

Step 1

Read the array.

Step 2

Read the target value.

Step 3

Use the first loop to select one element.

Step 4

Use the second loop to check every remaining element.

Step 5

If the sum of the two elements equals the target, print their indices.

Step 6

Stop the program.

#	LeetCode	Difficulty	Concept
<u>1</u>	Binary Search (#704)	Easy	Classic Binary Search
<u>2</u>	Search Insert Position (#35)	Easy	Lower Bound
<u>3</u>	Guess Number Higher or Lower (#374)	Easy	Binary Search
<u>4</u>	First Bad Version (#278)	Easy	Binary Search
<u>5</u>	Valid Perfect Square (#367)	Easy	Binary Search

BINARY SEARCH

Algorithm (Step-by-Step)

Step 1

Read the sorted array.

Step 2

Read the target element.

Step 3

Initialize two variables:

- `left = 0`
- `right = n - 1`

Step 4

Repeat while `left <= right`.

Step 5

Find the middle element.

`mid = (left + right) / 2`

Step 6

Compare the middle element with the target.

- If `nums[mid] == target`
 - Return `mid`.
- If `target < nums[mid]`
 - Search the left half.
 - `right = mid - 1`
- Else
 - Search the right half.
 - `left = mid + 1`

Step 7

If the loop ends without finding the target, return -1.

Basic string operations to interview-oriented string algorithms

Skill	Problems Covered
String Traversal	58, 125, 709
Character Manipulation	344, 709
Two Pointer Technique	344, 125
String Comparison	14

Searching in Strings	28
Character Frequency	242, 387
Hashing Basics	242, 387
String Rotation	796
Nested Loops	771
ASCII Concepts	709